GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

INSTRUCTIONS
GEK-41276A

DIGITAL TERMINAL
MAINTENANCE MODULE

PART NO. IC3607D202C

REV. 1 1/10/77	REV. 2	APPROVALS D. E. Morehart	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DIGITAL TERMINAL MAINTENANCE MODULE	
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REVISED INFORMATION

Textural information changed on the most recent revision is flagged by a vertical bar along the left hand margin.

I. DESCRIPTION OF EQUIPMENT

1-1. DESCRIPTION OF DIGITAL TERMINAL MAINTENANCE MODULE (DTMM). The Digital Terminal Maintenance Module (DTMM) is an optional, portable plug-in unit for off-line testing of on-line monitoring of digital control systems using inputs and outputs (I/O's) mounted in a Digital Remote Terminal (DRT). It is contained in a steel protective enclosure and may be plugged into any digital, I/O assembly. A front panel layout of the DTMM is shown in Figure 1-1. The DTMM is extremely helpful during the installation and test of control systems using a digital logic controller, such as a Directo-Matic* Digital Logic Controller (DLC), 4000-series process computer, etc.

1-2. DTMM CONNECTION TO REMOTE TERMINAL. An attached length of multi-conductor cable from tne console terminates with an edge connector. This connector can be plugged into slot F of Digital Remote Terminal Line Interface (DRTLI). A typical connection inside a Digital Remote Terminal Line Interface Module is shown in Figure 1-2. The earlier model digital I/O sticks (IC4516AA) include the VTA as a part of the DRTLI rather than as a separate module. Special instructions applicable to IC4516AA VTA connections are contained on sheet 14A.

1-3. DTMM CAPABILITIES. The DTMM is used to monitor all input and output termination signals to verify the operation of ITU/OTU circuit cards and associated interconnecting cables. In the SIMULATE NORM or AUX modes, it generates data normally received from the controller (DLC-Basic) or Process Computer and it can force the state of output signals or read the state of input signals. This allows the remote terminal to be completely disconnected from the basic controller and tested off-line. In new installations where the DRT's are received prior to the basic controller equipment, these terminals can be installed and tested prior to the arrival of the DLC Basic. Figure 1-3 shows the DTMM connected to a typical long line process control system.

1-4. DTMM PLUGABILITY USING DIGITAL REMOTE TERMINAL LINE INTERFACE ASSEMBLY (DRTLI). The DRTLI provides for on-line or off-line plugability of the DTMM by means of cables and jumper cards. The DRTLI interface slots are shown in Figure 1-4. When the DTMM selector switch is set to MONITOR, slots G and F are used to mechanically connect the data lines to the DTMM as follows:

A. Insertion of DTMM Cable into DRTLI for Monitor Mode.

1. Before attempting on-line plugability, ensure the DTMM function selector is set to MONITOR position.

2. If present, remove jumper card PWB68A947088G1 from slot G of DRTLT. If not present, obtain from DTMM for use in step 4.

3. Insert DTMM cable into slot F of DRTLI.

NOTE

Data lines are not connected to slot F of DRTLI until step 4 of this procedure has been performed. Only the P14V, DCOM, and SIMD signals are connected.

4. Install jumper card PWB68A947088G1 into slot G; this connects data lines to slot F.

* Trademark of General Electric Company, U.S.A.

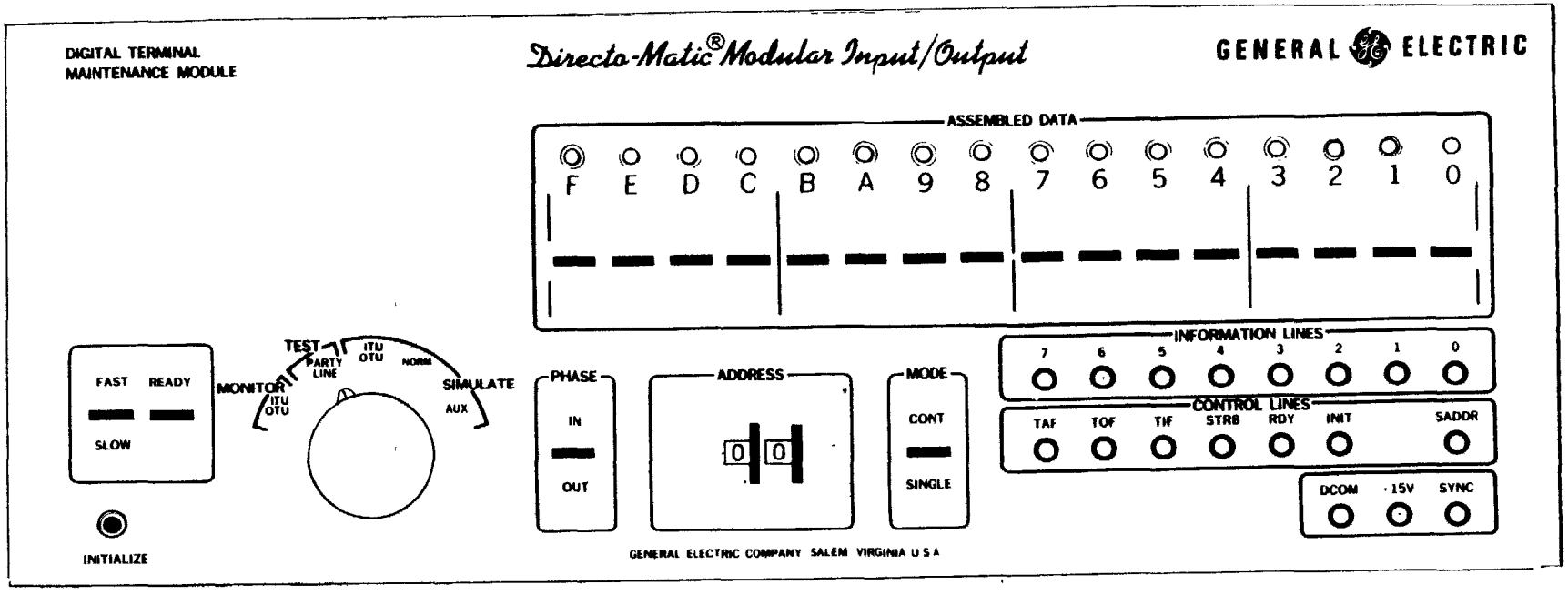


Figure 1-1. Front Panel Layout of DTMM

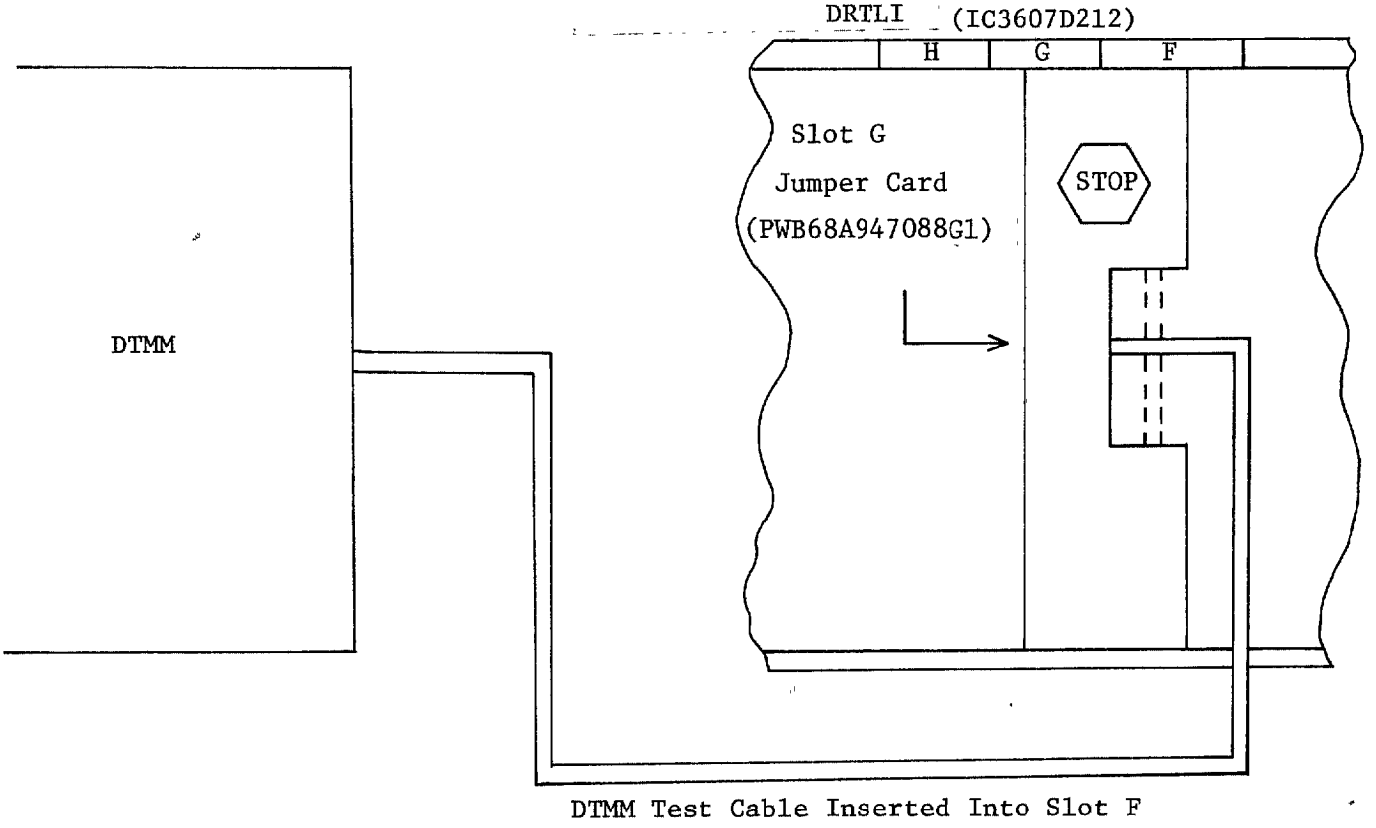


Figure 1-2. DTMM Connection to DRTLI Slot F of the Digital Remote Terminal Line Interface Module

B. Insertion of DTMM Cable into DRTLI for Simulate Normal Mode.

NOTE

The K cable disconnect (slot D) and jumper card PWB68A947088G2 (slot E) provide for isolating the entire DRTLI from the K cable for maintenance purposes or testing with the DTMM in the simulate mode. The mechanical interlock provision insures that the last data line to "break" upon pulling the jumper cards, and conversely the first data line to "make" upon insertion of both jumper cards will occur during the SIMD mode. This allows the IC3607D212 DRTLI to be on-line-plugable with the K cable regardless of the DTMM cable being present in slot F.

1. Remove jumper card PWB68A947088G2 from slot E.

2. Remove K cable jumper PWB (IC3600DJBB1) from slot D.

3. Set DTMM function selector to SIMULATE POSITION.

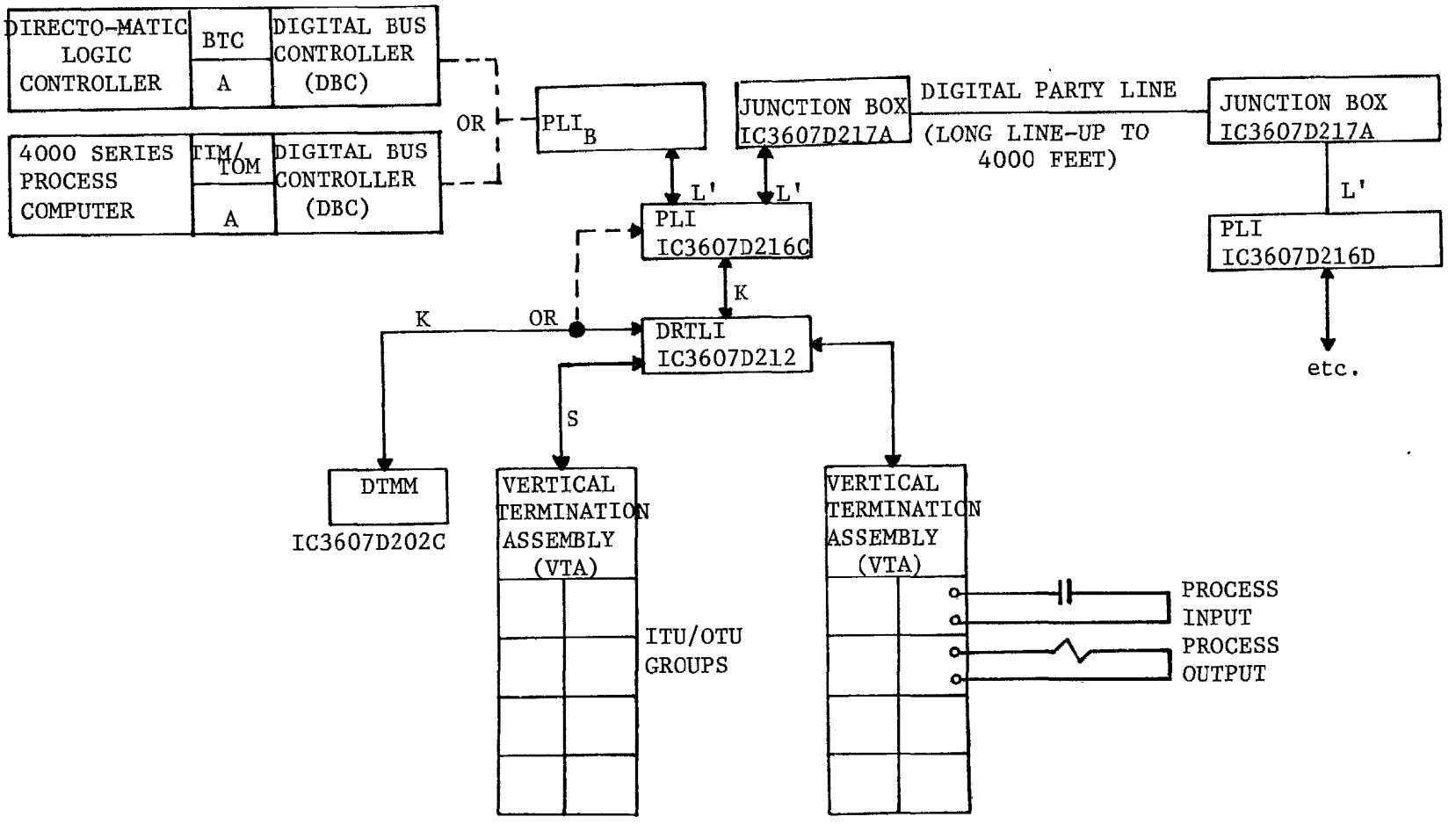


Figure 1-3. DTMM Connected to a Typical Long Line Process Control System

REV. 1 1/10/77
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ISSUED Feb. 2, 1976
PRINTS TO
MADE BY K. K. Cooper
APPROVALS D. E. Morehart
FIRST MADE FOR REQ
GENERAL ELECTRIC
SALEM, VA. U.S.A.
DIGITAL TERMINAL MAINTENANCE MODULE
277A4900
CONT. ON SH. 7
SH NO 6

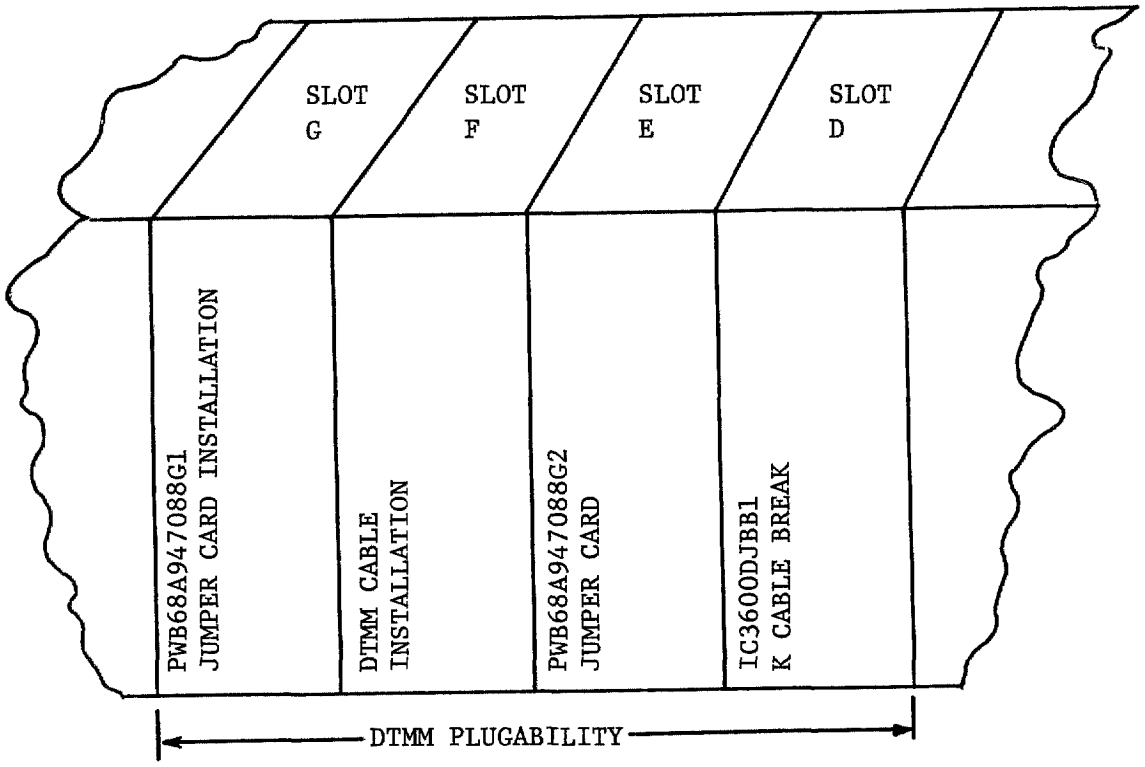


Figure 1-4. Part of IC3607D212 Digital Remote Terminal Line Interface Assembly (DRTLI)

NOTE

The slot F DTMM cable and slot G PWB68A947088G1 jumper card are stored inside the back of the DTMM. Slot F and G of the DRTLI assembly provides only the female half of the connector.

C. Return of DTMM to Monitor Mode. To change the DTMM from the simulate mode to the monitor mode, proceed as follows:

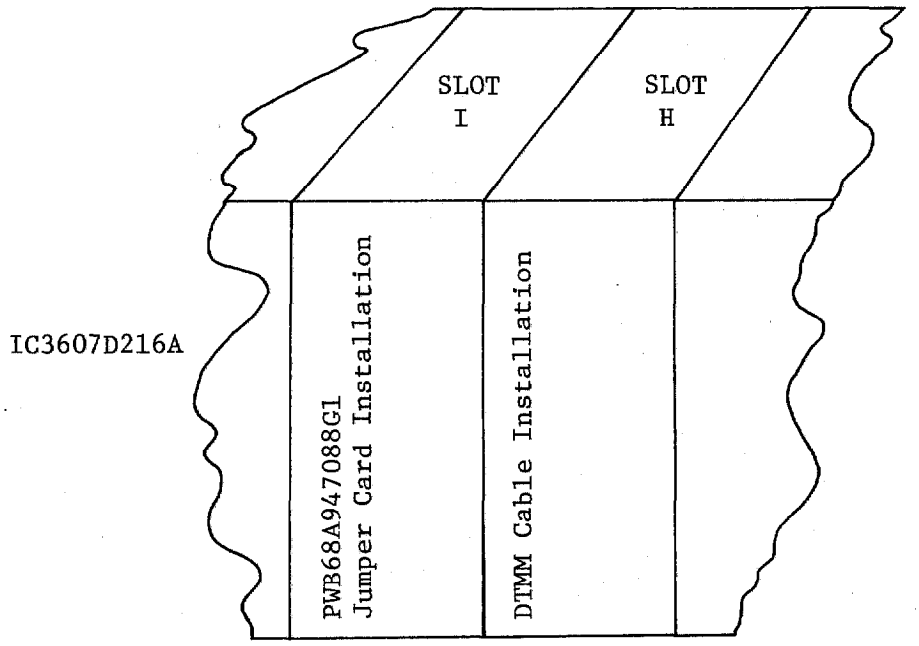
1. Set Function Selector switch to MONITOR position.
2. Insert K cable jumper PWB (IC3600DJBB1) into slot D.
3. Insert jumper card PWB68A97088G1 into slot E.

D. Removal of DTMM cable from DRTLI.

1. Set DTMM function selector to the MONITOR position.
2. Remove jumper card PWB68A97088G1 from slot G.
3. Remove DTMM cable from slot F and store in DTMM. slot G, and store return it to the DTMM storage slot.

1-5 DTMM PLUGABILITY USING PARTY LINE INTERFACE (PLI). The short line PLI (form A) and the long line (forms C and D) permit plug-in of DTMM while on-line, at receptacle H (form A) or receptacle D (forms C and D). See Figure 1-5, PLI Interface Connector Slots. The DTMM receptacle is mechanically interlocked with an adjacent jumper card (PWB68A947088G1). Upon plug-in of the DTMM cable, only the P14V, DCOM, and SIMD signals are applied. Upon reinsertion of jumper PWB68A947088G1, the data lines are reconnected.

Part of PLI between Processor and Digital I/O Short Line System



Part of PLI between long line and DRT line interface (IC3607D212).

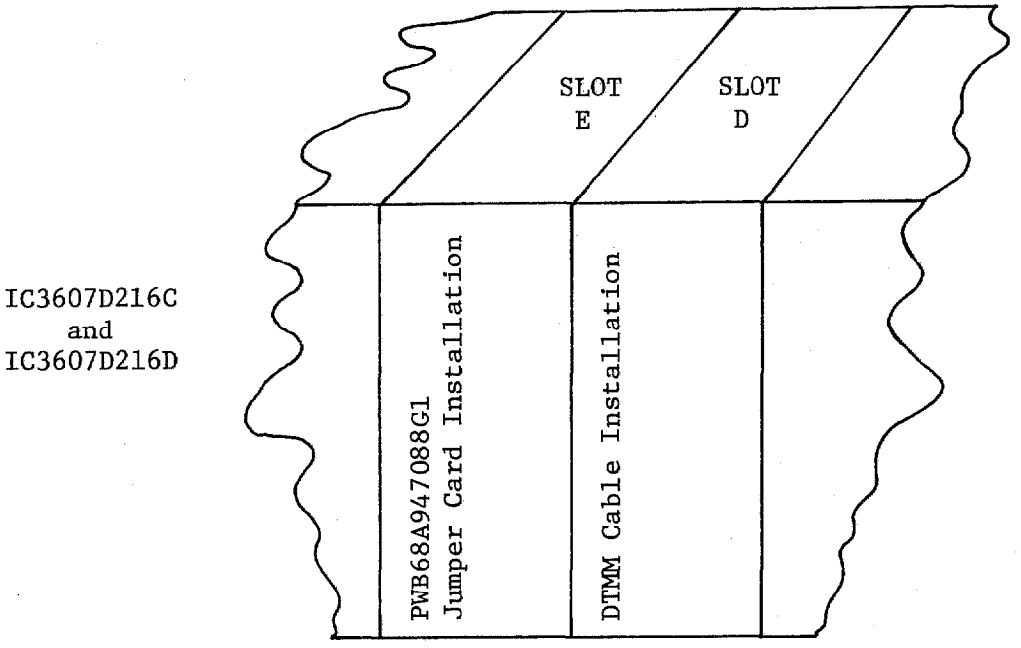


Figure 1-5. PLI Interface Connector Slots

NOTE

The DTMM is on-line plugable at the PLI for the monitoring function only. For the simulation mode, the system is taken off-line, the PLI power turned off, and the control card removed. The control card is located in slot C (PLI form A), or in slot H (PLI forms C and D).

CAUTION

Putting the Function Selector in a SIMULATE position without removing the control card may cause a system malfunction.

1-6. DTMM SWITCHES,LIGHTS AND TEST JACKS. The switches, lights, and test jacks mounted on the DTMM are shown in Figure 1-1 and function as follows:

A. INITIALIZE (Pushbutton). This pushbutton generates an internal initialization signal for the DTMM. It also generates the DRT initialization if the function is present in the DRTLI. This pushbutton is used during the Simulate NORM and AUX modes only.

B. PHASE (2-Position Switch). This switch is used in the Monitor, Simulate NORM, and Simulate AUX operating modes.

1. IN (Up Position) Selects the input data to be monitored or simulated. The ITU in question is checked by visually comparing the status of the data indicators on the console with the actual ITU inputs.

2. OUT (Down Position) Selects the output data to be monitored or simulated. The status of the console's data switches is sent to the DRTLI under test to check the proper response of the OTU.

C. MODE (2 Position Switch)

1. CONT (Up Position) The selected address is continuously monitored or generated.

2. SINGLE (Down Position, Spring Return to Center). When depressed, a single output or input cycle is sent to the selected address.

D. ADDRESS (Two Thumbwheels). Selects the I/O address to be monitored or tested.

E. FAST-SLOW (Two Position Switch) Used in Simulate NORM and Simulate AUX selector positions only.

1. FAST (Up Position) Selects the fast (60 microseconds) time frame of the digital bus controller (DBC).

2. SLOW (Down Position) Selects the slow (120 microseconds) time frame of the DBC.

F. READY (Two Position Switch)

1. READY (Up Position) Simulates the "ready" condition normally generated by the PLI. This switch position is used in SIMULATE NORM and AUX tests.

2. Down position simulates loss of strobe to PLI module (DLC stall).

G. ASSEMBLED DATA Switches (Sixteen Two Position Switches). Used in SIMULATE NORM and AUX tests when data is generated by the DTMM.

1. Up position generates a digital "zero" as viewed from INFORMATION LINES jacks. Outgoing cable data is inverted.
2. Down position generates a digital "one" as viewed from INFORMATION LINES jacks. Outgoing cable data is inverted.

H. ASSEMBLED DATA Indicators (Sixteen Lights). The use of the ASSEMBLED DATA indicators depends on the setting of the function selector. The indicators are operational in all four settings of the function selector.

I. Function Selector. (Five Position Rotary Switch).

1. MONITOR. In this setting, the DTMM is essentially a passive device and does not drive the data or control lines to which it is connected. The DTMM captures and displays the data transferred to or from the I/O address selected by the ADDRESS thumbwheels. This digital information is displayed on the indicators and updated each time the designated address is detected and the appropriate direction of data is detected. Outputs to the address selected will not affect the display when the PHASE switch is in the IN (up) position. Similarly, inputs from the address selected will not affect the display when the PHASE switch is in the OUT (down) position. An oscilloscope sync pulse of positive polarity is generated each time the display is updated. With the MODE switch set to the SINGLE (down) position, the data will only be updated on the first detected cycle after depressing the MODE switch. The sync pulse is also generated only once.

2. TEST (PLT). The TEST position is applicable when a diagnostic program is being run by the controlling processor. The diagnostic program initiates the transfer of data to the address set on the ADDRESS thumbwheels. The DTMM receives and stores this data. When the controlling processor initiates an input cycle from the same address, the DTMM responds by sending the stored data back through the system for verification. This response is an indication that the digital bus controller, digital party line, party line interface, and digital remote terminal line interface (DRTLI) are all functioning properly.

3. SIMULATE ITU OTU. This setting is used to test potential additions of data into the system. As such, the DTMM simulates a new ITU or OTU group; one which is not currently implemented within the system. The module responds to data outputs by lighting the ASSEMBLED DATA indicators as in the monitor mode, but responds to inputs by sending data corresponding to ASSEMBLED DATA switch positions. The data transmitted according to ASSEMBLED DATA switch positions is also displayed on the ASSEMBLED DATA indicators. The address of the simulated ITU or OTU is entered by the ADDRESS thumbwheels and the PHASE switch setting determines the direction of data flow. The PHASE IN switch setting causes the DTMM to simulate an ITU, and the PHASE

OUT setting allows simulation of an OTU.

CAUTION

Do not attempt to simulate an ITU address presently established in the system.

4. SIMULATE NORM. This setting is used with a DLC for testing ITU's and OTU's, and a DRTLI independently of the processor and digital party line. In the SIMULATE NORM mode, an address is designated by the ADDRESS thumbwheels and data input or output is selected by the PHASE Switch. The SIMULATE NORM setting allows generation of the control signals normally produced by the digital bus controller. If PHASE OUT is selected, the data on the ASSEMBLED DATA switches is sent to the address designated by the ADDRESS thumbwheels. The OTU can then be checked for proper response. When PHASE IN is selected, the specified ITU is checked by comparing the status of the ASSEMBLED DATA indicators on the DTMM front panel with the actual ITU inputs.

NOTE

For on-line plugability using the DRTLI, refer to paragraph 1-4.

5. SIMULATE AUX. The DTMM operation in this function selector setting is similar to SIMULATE NORM operation but is used on those systems under control of a series 4000 process computer. The process I/O is the same as that of the DLC system except for the digital bus controller which interfaces the digital party line. Input data is sent in groups of three 8-bit bytes (24 bits) instead of two (16 bits).

J. INFORMATION LINES (15 Test Jacks). These two horizontal rows of test jacks can be used in all function selector positions. They are oscilloscope monitoring jacks to display the status of equivalent DRTLI signal lines. The signal line to each jack in the console is buffered. Data from the INFORMATION LINES, CONTROL LINES and SYNC jack are available for monitoring. This data is applicable to any address on the digital party line.

K. DCOM, +15V, SYNC (Test Jacks). The +15V jack can be used to check the output voltage (normally +14VDC) of the DRT power supply. A positive sync pulse is generated at the SYNC jack prior to each address recognition.

II. TEST INSTRUCTIONS

2-1. DTMM OPERATING MODES. The DTMM function selector is used to select the type of test operation desired. The test modes are as follows:

A. MONITOR Operation. In the MONITOR position of the function selector and the MODE switch in the CONT (continuous) position, the DTMM provides an on-line display of both the input and output data on the entire party line system. This display, by means of the ASSEMBLED DATA indicator lights, provides confirmation that both the I/O's and the controller are operating correctly. A specific control group may have a unique signature that reflects normal operation. This observation is valuable to the experienced technician in determinging that something has changed, or does not look right. An on-line process controller exhibits input/output traffic which reflects the state of the control. In a regulator, this traffic includes reference, feedback and error functions. Data logging applications also receive the logged variables. The observation of this data flowing between the digital I/O's and the digital bus controller provides diagnostic information pertaining to both the controller and the process.

B. PARTY LINE TEST (DLC Only). The operational party line test requires Part Two of software documentation (IC3609DPLT1). All system outputs should be disabled since this test sends random patterns to all OTU addresses. To perform the party line test, proceed as follows:

1. Turn the DTMM function selector to PARTY LINE TEST position, the PHASE switch to IN, MODE switch to CONT, and FAST SLOW switch to FAST.

2. Set the ADDRESS thumbwheels to 00.

3. Load the test IC3609DPLT1 into the system scratch memory by means of Autoload. The test uses up to @FF of address space.

4. The test comes on immediately after loading.

5. The ABS switches select the test parameters. For a description of the ABS module refer to System Instruction Book GEK 40421.

SW0	SW1	Operating Speed
0	0	36 Microseconds/Frame
1	0	66 Microseconds/Frame
0	1	144 Microseconds/Frame
1	1	264 Microseconds/Frame

SW2 1 = Party Line Test No. 1, Test 5
SW3 1 = Party Line Test No. 2, Test 6

(1 designates switch ON or "up" position).

6. The test results appear on the ABS lamps. The number of the highest test passed flashes in the lights upon completion. (See Table 2-1).

Table 2-1. Party Line Test Results

ABS Indicators	Condition Detected
00XXX E3210	Test Passed, X flashes in pattern (0100), (0101) (0110).
10001	Failed Initialization Test
10010	Failed Strobe Inhibit Test
10011	Failed Register Test
10100	Failed Frame Time Test
10101	Failed PL Test No. 1
10110	Failed PL Test No. 2

Restart @ 20.

7. If the system includes various remote DRT's, move the DTMM to each of these in turn and run the test at these locations.

C. ITU/OTU SIMULATE Test. With the function selector set to the ITU/OTU position, outputs from the processor are displayed on the console's indicators. This is a special user function and only dedicated addresses are used. For OTU simulation, the output address is set in by the ADDRESS thumbwheels and the PHASE switch depressed to the OUT position. All output data sent to the selected address is displayed on the ASSEMBLED DATA indicators. For ITU simulation, the PHASE switch is in the IN (input) position, the input address is set in by the ADDRESS thumbwheels and data entered by means of the 16 ASSEMBLED DATA switches. The ASSEMBLED DATA indicators show the status of the data switches. This information is sent to the processor via the DRTLI and digital bus controller.

D. SIMULATE NORM Function. All VTA's containing ITU's and OTU's inside a digital remote terminal can be tested with the DTMM. Power must be available for the P14V power supply and the external circuit feeders. To perform the SIMULATE NORM function, proceed as follows:

1. Set DTMM function selector to the MONITOR position.
2. Remove jumper card PWB 68A947088G1 from slot G of DRTLI if present. This card is normally stored inside the DTMM.
3. Insert DTMM cable into slot F of DRTLI.
4. Install jumper card PWB 68A947088G1 into slot G; this connects data lines to slot F.
5. Remove jumper card PWB 68A947088G2 from slot E.
6. Remove K cable jumper PWB (IC3600DJBB1) from slot D.
7. Install PWB 68A947088G2 back into slot E.
8. Set DTMM function selector to SIMULATE NORM position.
9. Apply power to the DRT VTA. The P14V bus is adjusted by the regulator for 14 volts. Energize the feeders to power the external process driven devices.

10. Output Checks

(a) Set DTMM function selector to SIMULATE NORM position, PHASE switch to OUT, MODE switch to SINGLE, FAST SLOW switch to FAST, and the READY switch to the down position.

(b) Determine the address of the output group to be tested by observing the nameplate on the DRTLI. Set the group number (10, 3F, or similar) into the ADDRESS thumbwheels. Set READY switch to the READY (up) position and the MODE switch to the CONT position to start the outputs. Depress the ASSEMBLED DATA switches one at a time and the requested output should appear at the addressed OTU light. Also the associated ASSEMBLED DATA light should illuminate. In the case of a solenoid driver OTU, IC3606DSDA or IC3606DSDC, the OTU light goes off when the output appears.

(c) Set MODE switch to SINGLE position and turn the thumbwheels to the next address; the OTU should maintain the previous setup. Depress the READY switch and the outputs should clear. As the test proceeds, it is best to compare the operation of the driven devices to the DTMM ASSEMBLED DATA lights to verify correct operation and the integrity of associated wiring. The status of the OTU indicator lights should also match the ASSEMBLED DATA light display. Continue testing the VTA's until all outputs are checked. Figure 1-6B shows the timing relationships which should be observed at the CONTROL LINES jacks on the DTMM.

11. Input Checks

(a) Turn DTMM function selector to SIMULATE NORM position, PHASE switch to IN, MODE switch to CONT, and FAST SLOW switch to FAST.

(b) Determine the address of the input groups to be tested by observing the nameplate on the VTA. Set this group number (10, 3F, or similar) into the ADDRESS thumbwheels. Operate the inputs one at a time. The associated ASSEMBLED DATA light should come on as well as the light on the ITU thereby indicating an energized input. As the test proceeds, it is best to check the operation of the input devices as well as the ITU's to verify proper operation and the integrity of associated wiring. Figure 1-6C shows timing relationships which should be observed on the CONTROL LINES jacks on the DTMM.

E. SIMULATE AUX (4000 Series Tests). This test is identical to the SIMULATE NORM Function except that inputs and outputs are combined on the same time frame. The input portion of the frame incorporates 24 data bits whereas the output portion incorporates 16 data bits. All ITU's and OTU's for one frame are normally assigned the same address. Figure 1-6A shows the relationship of timing control pulses to address, output data (16 bits), and input data (24 bits) between the DBC and the DRTLI.

2-2 SPECIAL INSTRUCTIONS FOR USE OF IC4516AA VTA's (Early Models).

NOTE

When testing earlier I/O VTA's (IC4516AA), DTMM IC3607D202B must be used. When testing newer I/O VTA's (IC4516AC), DTMM IC3607D202B must be used with conversion cable(68A947079G02L120), and DTMM jumper card PWB68A947088G1.

A. DTMM CONNECTIONS TO ISOLATE ENTIRE REMOTE TERMINAL. To connect the DTMM to the remote terminal, proceed as follows:

1. Remove power from the digital remote terminal.
2. Plug the DTMM into the connector at the top of any VTA.
   (a) Remove protective cover from the top connector of the VTA.
   (b) Align the DTMM plug on the exposed pins and jack the plug into position with the jacking screws.
3. Isolate the DRT from the remainder of the party line system by removing the control card (IC3600DLCA or IC3606DLCB) from the PLI.
4. Reapply power to the DRT.

B. DTMM CONNECTIONS TO ISOLATE A SINGLE VTA. Single VTA's are connected as follows:

1. Remove power from the DRT.
2. Remove party line connectors from top of VTA while keeping them connected together. The last VTA in the lineup will have a single connector.
3. Insert DTMM plug into top of VTA backboard, then connect normal cabling on top of DTMM plug.

NOTE

Use of the above procedure isolates the VTA from the DRT signal flow while supplying power to the VTA. Therefore, the VTA can be operated without affecting the other VTA's on the remote terminal.

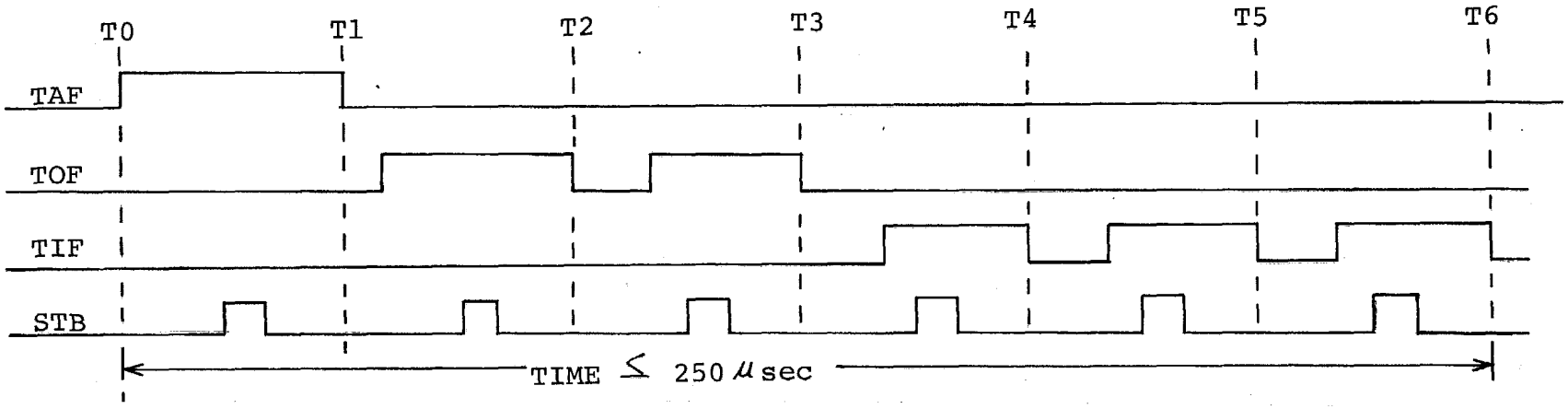
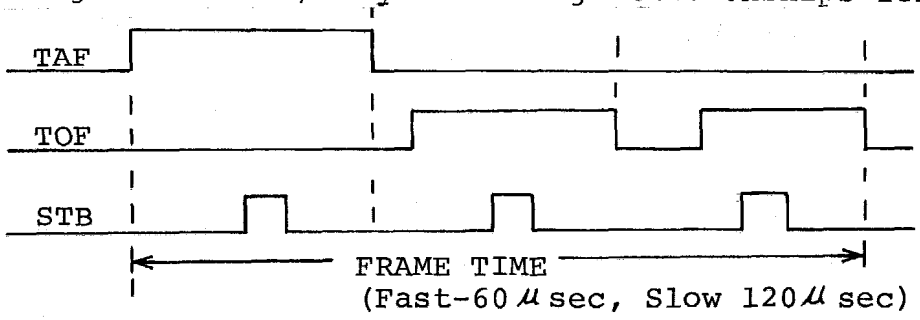


Figure 1-6A. I/O Cycle Timing Relationships for 4000 Series Processor



Note: All signals shown on this page are CONTROL LINES outputs from a DTMM front panel operating in Simulate mode.

Figure 1-6B. Output Cycle Timing Relationships for DLC Processor

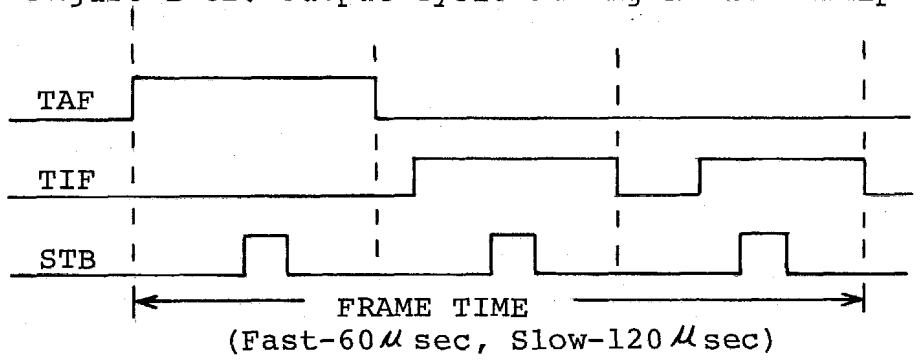


Figure 1-6C. Input Cycle Timing Relationships for DLC Processor

III. MAINTENANCE INSTRUCTIONS

3-1. DTMM MAINTENANCE. The DTMM is mounted in a steel enclosure and requires no special care during use or storage. Do not attempt to clean or repair switches. If a switch does not function properly, it should be replaced. Recommended environmental conditions are as follows:

A. Operating Temperature (ambient): Zero degrees C to 55 degrees C.

B. Storage Temperature: Minus 40 degrees C to plus 70 degrees C.

C. Humidity (operating and storage): 95 percent without condensation.

3-2. DTMM STORAGE. When not in use, the following steps should be performed:

A. Store the DTMM cable with attached connector in the slot provided at the back of the DTMM enclosure.

B. Store PWB 68A947088G1 jumper card in slot provided at back of the DTMM card rack.

C. Leave the ASSEMBLED DATA switches and other spring-loaded switches in the center or NULL position.

D. Leave the function selector switch in the MONITOR position.

E. Select a storage area free from falling objects and highly toxic chemical gases. Do not store it for extended periods on a cluttered workbench in a shop location.

3-3. USEFUL PART (DRAWING NUMBERS):

DTMM IC3607D202B - (earlier cable)
DTMM IC3607D202C - (later cable)
Party Line Interface (PLI) IC3607D216
Digital Remote Terminal Line Interface DRTLI IC3607D212
Data Card IC3606DDB1, Sh. 3.0-3.6
Control Card IC3606DDCB1, Sh. 3.0-3.8
Module Elementary 68C946085, Sh. 3.0
Software Test Instructions IC3609DPLT1, Sh. 9.0-9.3
Conversion cable to connect from IC3607D202B to new PLI or DRTLI 68A947079G02L120

PWB68A947088G1-DTMM (or I&MM) Jumper Card DRTLI Jumper Card
PWB68A947088G2-K Cable Break Jumper Card